Cloudy with a Chance of Compute

Mapping the hidden geography of AI compute across 8,000 cities — and who it leaves behind

Matthew Scott Tan
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Introduction

Singapore sits 4 km from its nearest major cloud compute region. Lagos, one of the largest and fastest-growing cities on the African continent, sits nearly 3,800 km from its nearest. Both cities have significant scale, growing digital economies, and stated national ambitions in artificial intelligence. But from an infrastructure standpoint, they do not start from the same place.

Modern day AI systems depend on physical cloud compute infrastructure to train models, serve real-time inference, and support the applications that drive AI adoption. In particular, the main players in the AI industry, Amazon, Microsoft and Google, operate data centers under Amazon Web Services (AWS), Azure and Google Cloud respectively. These data centers are found in clusters in regions, concentrated in specific locations, primarily in North America, Europe, and parts of East Asia. This creates some geospatial meaning to the accessibility to AI, as the distance to these data centers affects network latency, data transfer costs, and the practical viability of running AI workloads at scale. Industry research estimates that network-induced latency alone adds 50 to 150 milliseconds per request for cities far from centralized cloud infrastructure, frequently exceeding the thresholds required for responsive AI applications.

The scale of this imbalance is significant and widening. In 2026, global hyperscaler capital expenditure on data center construction is projected to exceed $600 billion [1, 2], yet the distribution of that investment remains deeply uneven. Across the entire African continent, less than 1 percent of global data center capacity currently resides. Only South Africa hosts major cloud regions from all three leading providers. No AWS, Azure, or Google Cloud region currently operates in Nigeria, in East Africa's largest economies, or anywhere in North or West Africa. A population larger than that of Europe or the Americas is served by a single country's compute footprint. The International Monetary Fund has identified access to computing power and data infrastructure as a critical determinant of which countries benefit from AI [3, 4]. Furthermore, recent assessments indicate that AI adoption in the Global North is growing nearly twice as fast as in the Global South: a gap that multiple international bodies have warned risks deepening cross-country inequality rather than narrowing it [5, 6, 7]

Despite the growing recognition of this expanding divide, public discussion of AI advantage continues to emphasize talent pipelines, venture capital, and national policy frameworks. There is a lack of attention to the physical infrastructure layer, where compute capacity is actually located, how far cities sit from it, and whether that distance corresponds with observable differences in AI activity. There is, to date, no systematic global mapping of how cloud compute proximity relates to the geographic distribution of AI research.

This atlas seeks to address that gap. It measures the distance from each of the world's 8,000 largest cities to its nearest major cloud compute region, overlays observed AI research activity drawn from the OpenAlex scholarly record, and investigates a central question:

Once city size and geography are accounted for, do cities that are farther from cloud compute infrastructure tend to show less observed AI research activity, and if so, where are those disparities most consequential?

The interactive map below shows the starting point of the atlas: 8,000 cities, each positioned by its population and colored by its distance to the nearest major AWS, Azure, or Google Cloud region. Diamonds mark the cloud regions themselves, differentiating the different providers by color. The concentration is immediately visible from the patterns of the diamonds: dense compute corridors run through North America, Western Europe, and East Asia. Meanwhile, large stretches of Africa, Central Asia, and parts of Latin America sit far from any major cloud region.

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8,000 cities sized by population and colored by distance to the nearest major cloud compute region. Diamonds mark deployed AWS, Azure, and Google Cloud regions. Data: Natural Earth, hyperscaler public documentation. Built and published using ArcGIS Online.

The sections that follow describe how this atlas was constructed, what the resulting patterns reveal, and why the relationship between compute proximity and observed AI activity is more structured than it first appears. Beforee we begin, we first carry out someexploratory data analysis to understand the context.

How the atlas works

The atlas is built in three stages using ArcGIS Online and ArcGIS Pro, with supporting data processing in Python.

Stage 1: A global comparison frame. We start the analysis globally, looking at the world's 8,000 largest cities by population, drawn from the Natural Earth populated places dataset. This broad scope makes it possible to compare AI-linked cities to the wider global city system rather than focusing only on a small set of well-known technology hubs. Each city is geocoded with coordinates and population estimates.

Stage 2: Mapping cloud compute infrastructure and measuring access. The atlas maps 60+ major deployed cloud compute regions operated by the three leading hyperscalers, AWS, Azure, and Google Cloud, compiled from each provider's public infrastructure documentation as of early 2026. For each city, the geodesic great-circle distance to its nearest cloud region is calculated, producing a single, transparent, and globally comparable measure of compute accessibility. This distance metric is intentionally simple: it is a proxy for geographic proximity, not a full model of latency, pricing, or GPU availability. The purpose of this is to allow for comparability across thousands of cities using a consistent methodology.

Stage 3: Overlaying observed AI research activity and testing the relationship. The final stage overlays observed AI-related research activity derived from the OpenAlex scholarly record, matching recent AI publications (between 2020–2025) to cities by institutional affiliation. This produces a subset of 328 AI-linked cities within the 8,000-city frame. The atlas then tests whether the distance-activity pattern is statistically structured rather than anecdotal, using three methods: Getis-Ord Gi* hot spot analysis to identify significant spatial clusters of AI activity, Global Moran's I to measure spatial autocorrelation, and two spatial regression specifications (a Gaussian Process model and a conditional autoregressive model) to check whether the negative relationship between distance and AI activity survives controls for city population and geographic structure.

328 matched AI cities from the OpenAlex scholarly record, sized by recent AI works and colored by distance to the nearest cloud region. The dense cluster across Western Europe and East Asia — and the near-absence of points across Africa and Central Asia — mirrors the access gradient visible in the hero map above. OpenAlex matches are aggregated to unique cities. Built using ArcGIS Online.

When observed AI research activity is overlaid on the accessibility map, the correspondence is notable. Of the 328 cities matched to recent AI publications in the OpenAlex scholarly record (2020–2025, matched by institutional affiliation), 72 percent sit within 500 km of a major cloud region, while just 44 percent of all large cities in the 8,000-city frame. The median AI-linked city is 237 km from its nearest cloud region, while the median large city is 657 km away. The map above shows that AI research activity is not distributed in proportion to urban population, but rather is concentrated in the compute-proximate parts of the global city system.

What the Atlas Found

Finding 1: AI-linked cities occupy a fundamentally different part of the compute-access landscape.

The histogram below compares the distance-to-cloud distributions of two populations: all 8,000 large cities in the global frame (blue shading) and the 328 cities matched to recent AI research activity in the OpenAlex record (orange line). Both are plotted on a log-distance scale so that the full global range remains visible while the sharp leftward shift in the AI-linked sample is still clearly resolved.

Distance-to-cloud distributions for all large cities (n = 8,000, blue shading) and unique matched AI cities (n = 328, orange line) on a log-distance axis. Dashed vertical lines mark the respective medians: 237 km for AI-linked cities, 657 km for all large cities. Equal-width bands on a log scale. Source: Natural Earth city frame, OpenAlex AI-city match.

The separation is immediate obvious. The AI-linked distribution peaks sharply between 50 and 250 km, while the broader city distribution spreads across a much wider range, extending well past 2,500 km. To confirm that this visual separation reflects a genuine distributional difference rather than an artifact of sample size, three non-parametric tests were applied.

A two-sample Kolmogorov-Smirnov test was run, which measures the maximum vertical distance between two cumulative distribution functions. The results were that $D = 0.30$ ($p < 0.001$), confirming that the AI-linked and all-city distance distributions are drawn from different underlying populations. A one-sided Mann-Whitney U test was also implemented. This tests whether one distribution is systematically shifted below the other. The results confirmed that AI-linked cities are significantly closer to cloud regions than the broader city population ($U = 786,844$, $p < 0.001$). The effect is substantively meaningful, not merely statistically detectable, since by Cohen's d came out to be 0.74, corresponding to a medium-to-large effect size by conventional thresholds.

In concrete terms, 72 percent of AI-linked cities fall within 500 km of a major cloud region, compared to just 44 percent of all large cities. A chi-square test of independence on this 500 km threshold rejects the null hypothesis of equal proportions ($\chi^2 = 103.8$, df = 1, $p < 0.001$). Cities producing visible AI research do not simply sit somewhat closer to compute infrastructure on average. They concentrate inside the compute corridors, and are largely absent from the zones farthest from cloud regions.

Finding 2: The concentration sharpens when weighted by research volume.

Finding 1 treats each AI-linked city equally: a city with a single matched publication counts the same as one with hundreds. When the distribution is instead weighted by the number of recent AI works each city produces, the trend shifts further toward cloud infrastructure. The activity-weighted median distance drops to 164 km, compared to 237 km in the unweighted city-level view.

The intensification is visible in the concentration ratios. Within 500 km of a major cloud region, 72 percent of AI-linked cities account for 81 percent of all observed AI works, an excess concentration of approximately 9 percentage points over what would be expected if research output were distributed uniformly across the AI-linked sample. Within 250 km, 52 percent of cities produce 62 percent of all works. The cumulative share line in the figure below illustrates this acceleration: it reaches 81 percent at 500 km and 93 percent at 1,000 km.

Activity-weighted distance distribution. Bars show the share of recent AI works (not cities) in each log-distance band; the teal cumulative line tracks total concentration. Dashed line marks the weighted median at 164 km. Within 500 km, 72% of AI cities account for 81% of all observed works.

It is important to note what this finding does and does not show. It demonstrates that the cities with the highest research volume are disproportionately located near cloud infrastructure. A Spearman rank correlation between distance and research output within the AI-linked sample alone is weakly negative (correlation = −0.05, $p = 0.40$), indicating that once a city is already in the AI-producing set, additional proximity does not strongly predict higher output. The sharper concentration in the weighted view is driven primarily by the fact that the largest AI-producing hubs, cities with hundreds of matched publications, tend to coincide with the world's densest compute corridors, instead of by a smooth gradient within the AI-linked sample.

Finding 3: The pattern is spatially structured, not randomly distributed.

Findings 1 and 2 establish that AI-linked cities sit closer to cloud infrastructure than the broader city system. But a distributional difference alone does not tell us whether the pattern has *spatial structure* — whether nearby cities tend to share similar levels of AI activity, or whether high- and low-activity cities are scattered randomly across the map. Answering this question requires spatial autocorrelation analysis: a family of methods designed to test whether values observed at one location are statistically related to values at nearby locations.

Two complementary spatial diagnostics are applied here. The first, Global Moran's I, tests for the presence and strength of spatial autocorrelation across the entire matched-city network. The second, Getis-Ord Gi*, identifies *where* statistically significant clusters of high or low values occur. Together, they answer two distinct questions: is the pattern structured (Moran's I), and where exactly are the clusters (Gi*)?

Global Moran's I is a summary statistic that measures the degree to which a variable (in this case, log AI works) is spatially autocorrelated across a set of observations. It compares each city's deviation from the global mean against the deviations of its spatial neighbors, weighted by a spatial weights matrix. A positive Moran's I indicates that similar values tend to cluster together (high near high, low near low). A value near zero indicates spatial randomness. A negative value indicates dispersion. Applied to the 319 unique matched AI cities, the test returns $I = 0.066$ with a permutation z-score of 2.86 ($p = 0.008$). The autocorrelation is modest in magnitude but statistically significant: cities with high AI activity tend to sit near other high-activity cities, and low-activity cities cluster together as well. The pattern is not random.

Getis-Ord Gi hot spot analysis* moves from the global summary to local identification. While Moran's I tells us that spatial structure exists, it does not tell us where. Gi* computes a local statistic for each city by comparing the sum of AI works in its neighborhood against what would be expected under spatial randomness. Cities where the local sum is significantly higher than expected are classified as hot spots; cities where it is significantly lower are classified as cold spots. Significance is assessed at the 95 and 99 percent confidence levels using z-scores derived from the spatial weights structure. Applied to the matched-city network using ArcGIS spatial statistics tools, the analysis identifies 7 statistically significant hot spots and 33 cold spots. The hot spots cluster in East and Southeast Asia — with Macau as the single 99-percent hot spot — while cold spots concentrate in the Americas and peripheral Europe, including Carbondale (United States) as the strongest cold spot.

Getis-Ord Gi* cluster classification for 319 unique matched AI cities. Color indicates cluster class: hot spot at 99% (1 city), hot spot at 95% (6 cities), not significant (279 cities), cold spot at 95% (23 cities), cold spot at 99% (10 cities). Marker size scales with recent AI works. Global Moran's I = 0.066, z = 2.86, p = 0.008.

We chose these methods for several reason. Spatial autocorrelation diagnostics are not commonly applied to AI infrastructure or research-output data — most studies in this domain rely on descriptive mapping, regression, or network analysis alone. Moran's I and Gi* were selected here for three reasons. First, they are specifically designed to detect whether a spatial pattern is statistically structured or merely visually suggestive — a critical distinction for a project that covers 8,000 cities across diverse geographies. Second, they operate at two complementary scales: Moran's I provides a single whole-map test of spatial dependence, while Gi* provides city-level cluster assignments that can be mapped and interpreted individually. Third, both methods are implemented in the ArcGIS Spatial Statistics toolbox, making them reproducible within the Esri ecosystem that this project is built on.

Alternative approaches such as LISA or kernel density estimation were considered but Gi* was selected for its more interpretable hot/cold classification for a public-interest audience.

Finding 4: The atlas identifies 1,988 priority cities where compute distance and absent AI activity converge.

Findings 1–3 show that AI-linked cities are disproportionately close to cloud infrastructure and that this pattern is spatially structured. The natural follow-up question is: which cities are on the wrong side of both gaps, i.e. far from compute **and** absent from the AI research overlay?

To answer this, the atlas applies a screening rule. A city qualifies as a priority case if it meets two conditions simultaneously: zero observed AI works in the delivered OpenAlex overlay, and a distance to the nearest cloud region that exceeds the upper-quartile threshold of 1,252 km. Cities meeting both criteria are then ranked by population first and compute distance second, so that the largest underserved cities surface at the top of the priority list.

The screen flags 1,988 cities across 125 countries. The findings show that the concentration is not random and maps onto the same geographic structure identified in Findings 1–3. Priority cities cluster heavily in Sub-Saharan Africa, where 468 cities qualify (with a median compute distance of 2,676 km and only 3 cloud regions on the entire continent, all in South Africa), and in Latin America, where 704 cities qualify (median distance 1,298 km, with cloud infrastructure concentrated in Brazil and Chile). Smaller but significant clusters appear in Central Asia, the Middle East, and parts of Southeast Asia.

Notably, the top-ranked priority cities are not marginal settlements, but rather include some of the world's fastest-growing urban centers and cities with millions of residents, active digital economies, and declared interest in technology and innovation. Lagos, Kinshasa, Khartoum, Lima, and Bogotá all appear in the top ranks. They are places where the gap between urban scale and compute accessibility may be most consequential for AI participation.

This screening layer transforms the atlas from a descriptive exercise into a practical public-interest tool. For planners, innovation agencies, and international development organizations, the priority list identifies where infrastructure investment in cloud compute could address the starkest mismatches between urban demand and current supply. It is not a causal forecast — it does not predict which cities *will* develop AI activity — but it highlights where structural infrastructure conditions are currently least favorable.

Priority-city screening layer. Cities qualify when observed AI works are zero and distance exceeds the upper-quartile threshold (1,252 km). The top 100 are emphasized. Color encodes priority rank (1 = highest); marker size scales with population among the top 100. 1,988 cities across 125 countries meet the screening criteria. Ranking sorts by population first, then compute distance. This is a policy-screening layer, not a causal forecast.

Finding 5: The distance-activity relationship survives spatial regression controls.

Findings 1–4 show that AI-linked cities are closer to cloud infrastructure, that the concentration sharpens with research volume, that the pattern is spatially structured, and that nearly 2,000 cities sit on the wrong side of both gaps. But none of these findings account for the possibility that the distance-activity pattern is driven entirely by city size or broad continental geography. Logically, a large city is more likely to host AI research and more likely to attract a nearby cloud region. This means population could be a confounding variable in this natural experiment. Similarly, cities in Europe and North America may score high on both dimensions simply because of their continental position, not because of a direct distance-activity link.

To test whether the relationship between compute distance and AI activity holds after controlling for these factors, two spatial regression models were estimated: a Gaussian process (GP) model and a conditional autoregressive (CAR/GMRF) model. Both regress log AI works on two predictors: log distance to the nearest cloud region and log city population. The key difference between the methods lies in how they handle spatial dependence, which we view as the tendency for nearby cities to have correlated outcomes regardless of their individual characteristics.

The **Gaussian Process model** treats spatial dependence as a smooth, continuous surface. It estimates a latent spatial field across all city locations, allowing the model to absorb broad regional gradients, such as the general tendency for European cities to have both high access and high AI output, before estimating the residual effect of distance and population.

The Gaussian Process specification returns a distance coefficient of −**0.207** (per 1,000 km) and a population coefficient of +**0.279** (log scale). The negative distance coefficient means that, even after accounting for population and the smooth spatial field, cities farther from cloud infrastructure tend to produce less observed AI research.

The **conditional autoregressive (CAR/GMRF) model** handles spatial dependence differently. Rather than estimating a smooth surface, it models spatial correlation through a neighborhood structure, i.e. each city's outcome is partially determined by the outcomes of its spatial neighbors. This is a more aggressive spatial control, as it absorbs not only broad regional gradients but also local spillover effects between nearby cities. Under this specification, the distance coefficient comes out to −**0.052**, while the population coefficient remains positive at +**0.309**. The results are expected and informative: the CAR model absorbs more of the distance signal into the spatial structure, but the sign does not flip. Distance to cloud infrastructure remains negatively associated with AI research output even under the more demanding specification.

The notabele feature across both models is **consistency in direction**: the distance coefficient is negative in both, and the population coefficient is positive in both. This provides a sign check, and is not to be mistaken for a causal effect estimate. The current pipeline stores posterior point estimates but not the full uncertainty intervals needed for formal inferential comparison. Intsead, these models confirm that the descriptive pattern identified in Findings 1–4 is not an artifact of city size or broad geography. The trends persist as a structured spatial relationship once these factors are controlled for.

This is why the atlas treats compute accessibility as a meaningful spatial correlate of observed AI activity, rather than as a proven causal mechanism. The project does not claim that relocating a city closer to a cloud region would mechanically increase its AI research output. It claims that distance to compute infrastructure is one observable dimension of a broader geographic structure that corresponds with where AI activity is — and is not — currently concentrated.

Point estimates from two spatial regression specifications: Gaussian process (GP) and conditional autoregressive (CAR/GMRF). Distance to nearest cloud region (per 1,000 km) is negative in both models (GP: −0.207, CAR: −0.052); log population is positive in both (GP: +0.279, CAR: +0.309). The attenuation from GP to CAR reflects more aggressive spatial dependence absorption. Point estimates only — interval estimates are not available in the current pipeline outputs. These are directional sign checks, not causal effect sizes.

Beyond Distance: The Infrastructure Bundle

Distance to the nearest cloud region is the simplest access measure in the atlas, but it is not the whole story. Cities differ not only in compute proximity, but also in the wider infrastructure bundle that surrounds that proximity.

To capture that wider context, the atlas constructs a Compute Opportunity Bundle Index that scores each city on five components. The weights reflect a simple prior: physical proximity to compute is the most direct constraint on AI workload performance, so it receives the largest share. The remaining four components are weighted equally as enabling conditions that determine whether proximity translates into realized opportunity.

- **Proximity** to the nearest cloud region (40%): the core access measure and the atlas's primary variable.
- **Provider diversity** represents the distinct hyperscalers reachable within threshold distance (15%): captures whether a city depends on a single provider or has competitive options.
- **Redundancy** is represented by the total cloud regions within reach regardless of provider (15%) : measures resilience and capacity depth.
- **Urban scale or** city population (15%): proxies for market depth, labor availability, and demand concentration.

- **Institutional depth** or presence of top-ranked AI research institutions from the OpenAlex overlay (15%): captures whether the local knowledge ecosystem supports AI activity.

Each component is normalized to a 0–100 scale and combined as a weighted sum.

The map below applies the bundle index to the top 1,000 cities by population. Color encodes the composite score on a 0–100 scale: yellow marks cities with the strongest overall bundle (high proximity, multiple providers, deep institutions), while purple marks cities where the infrastructure ecosystem is thinnest. Additionally, the marker size scales with population. The geography is similar but not identical to the distance-only view in the main map. Western Europe and the US Northeast remain strong have strong compute, but East Asian cities rise further once institutional depth and provider density are factored in. Parts of the Middle East that scored well on distance alone drop visibly, reflecting proximity to cloud regions without the surrounding ecosystem to match. Sub-Saharan Africa and Central Asia remain consistently low on both measures, confirming the disadvantaged pattern from Finding 4.

Compute Opportunity Bundle Index applied to the top 1,000 cities by population. Color encodes the composite score (0–100); marker size scales with population.

The scatter plot below makes the bundle's analytical contribution concrete. Each dot is one of the top 1,000 cities by population. The x-axis shows the distance-only compute access score, representing how close a city sits to its nearest cloud region, on a 0–100 scale. The y-axis shows the full bundle score, representing the weighted composite accounting for the five criteria: proximity, provider diversity, redundancy, urban scale, and institutional depth. If distance were the only factor that mattered, every city would fall along the diagonal: high access would always mean high opportunity, and low access would always mean low opportunity.

The scatter plot reveals three distinct zones in the global city system. In the top right, cities like Paris, Tokyo, and London combine strong compute proximity with deep surrounding infrastructure. These are places where the full bundle reinforces itself. In the bottom left, cities across the Sahel, Central Asia, and interior South America score low on both dimensions, confirming the stacked-disadvantage pattern identified in Finding 4. But the most insightful zone is the middle: cities that score high on one dimension but not the other. Riyadh and Doha sit near the right edge of the x-axis, close to cloud infrastructure, yet drop sharply on the bundle score, reflecting proximity without the institutional ecosystem to match. Conversely, cities like Pune and Shenzhen sit lower on raw distance but climb on the bundle, pulled up by strong universities, large labor markets, and growing provider presence. These departures from the diagonal are exactly what the bundle is designed to surface. They represent cities where distance alone either overstates or understates the actual opportunity landscape, and they are invisible in a distance-only analysis.

This is why the atlas does not limit itself to an analysis of only distance. The bundle layer surfaces the cities where compute access is most likely to translate into realized AI opportunity — and the cities where proximity to a cloud region may be necessary but not sufficient.

Bundle score vs. distance-only access score for the top 1,000 cities by population. Dot size scales with population. Vertical spread at a given x-value reveals where provider diversity, redundancy, institutional depth, and urban scale push cities above or below their distance-only position. Top 1,000 cities by population; color = bundle score, size = population.

The bar chart below ranks the top 15 cities by overall bundle score. The list confirms what the scatter plot suggests: the strongest cities combine proximity with depth across multiple infrastructure dimensions. Paris, Tokyo, London, and Beijing lead — all four sit near major cloud regions, host world-leading AI research institutions, and are served by all three hyperscalers. But the ranking also surfaces cities that a distance-only analysis would miss. Pune ranks in the top 15 despite sitting farther from cloud regions than Gulf capitals, because its institutional base (home to multiple top-50 AI research institutions in the OpenAlex record) and large urban market compensate for the distance gap. Singapore appears in the top 15 consistent with its role as the atlas's clearest case of full bundle alignment, explored in detail below.

Top 15 cities by Compute Opportunity Bundle Index score (0–100). Rankings reflect the weighted composite of proximity (40%), provider diversity (15%), redundancy (15%), urban scale (15%), and institutional depth (15%). Cities that appear here but would not top a distance-only ranking illustrate the bundle's analytical contribution. Sources: Natural Earth, hyperscaler documentation, OpenAlex.

The global patterns are clear: compute access is geographically uneven, AI activity concentrates inside the densest compute corridors, and the relationship is spatially structured rather than random. But global patterns do not explain individual cities. The bundle helps by distinguishing cities with similar distance values but very different surrounding ecosystems. However, the bundle still just provides a composite score, not a narrative. To understand why some cities align with the atlas pattern while others depart from it, the analysis needs to move from indices to places. The four cases that follow do exactly that. Each represents one quadrant of the distance-activity relationship: a city where the full bundle aligns, a city where the research overlay understates a broader AI ecosystem, a city that outperforms what distance alone would predict, and a city where infrastructure constraints stack together. Together, they test the atlas against the complexity of real urban systems.

Four cities that explain the pattern

The atlas has established three things statistically. First, AI-linked cities occupy a fundamentally different part of the compute-access landscape than the broader city system — they are closer, and the difference is not an artifact of sample size (Finding 1). Second, that concentration sharpens when weighted by research volume: the highest-output cities sit disproportionately inside the densest compute corridors (Finding 2). Third, the pattern is spatially structured — not randomly scattered — with identifiable hot spots in East and Southeast Asia and cold spots across the Americas and peripheral Europe (Finding 3). A spatial regression confirms the relationship survives controls for city size and broad geography (Finding 5), and a priority screening layer identifies nearly 2,000 cities on the wrong side of both gaps (Finding 4).

But statistical patterns describe populations, not places. A Moran's I of 0.066 tells us that spatial clustering exists; it does not tell us what clustering looks like inside any particular city. A negative distance coefficient tells us the direction of the relationship; it does not explain why some cities beat the pattern while others conform to it. The bundle index begins to answer that question — it shows where compute access is reinforced or undermined by surrounding infrastructure — but even a composite score is a number, not a narrative. To understand how the atlas's global patterns actually play out, the analysis needs to move from indices to places.

The four cities that follow were chosen to represent the four quadrants of the distance-activity relationship, as analytical cases selected to test the atlas against the complexity of real urban systems:

Singapore: near compute, high AI activity

Dublin: near compute, lower AI than the infrastructure density suggests.

Ho Chi Minh Cit: farther from compute, high AI activity.

Lagos: far from compute, absent from the AI overlay.

Singapore

Distance: 4.1 km, Providers within 1,000 km: 3, Cloud regions within 1,000 km: 5, OpenAlex AI works: 1,07, Bundle score: 85.7/100

Singapore is the strongest confirmation case. It sits 4.1 km from its nearest cloud region (Azure Southeast Asia), has all three hyperscalers within 500 km, and scores far above the global reference on every bundle component.

Singapore sits inside the densest compute corridor in Southeast Asia. AWS, Azure, and GCP all operate within the city-state, with additional regions in nearby Kuala Lumpur and Jakarta extending the corridor south. SGIX, one of Asia's largest open and neutral internet exchanges, adds a network layer that connects Singapore into broader regional and global routing infrastructure. The city is not just close to cloud regions, it is embedded in a wider connectivity system that extends well beyond its small physical size.

At the local scale, cloud infrastructure, institutions, and national strategy reinforce one another. Singapore's National AI Strategy 2.0 treats AI as a national capability-building agenda [8], while the government's Data Centre Call for Application and Green Data Centre Roadmap manage capacity growth under energy constraints [9, 10]. The NUS Artificial Intelligence Institute, established in 2024, positions NUS as a global AI hub [12], while NTU's Artificial and Augmented Intelligence cluster anchors a long-established research ecosystem with deep corporate lab ties [13]. SGIX, one of Asia's largest open and neutral internet exchanges, adds the network layer [11]. The result is not just proximity to compute, but a broader system in which multiple enabling layers reinforce one another by design

Dublin

Distance: 0.0 km, Providers within 1,000 km: 3, Cloud regions within 1,000 km: 13, OpenAlex AI works: 179, Bundle score: 85.3/100

Dublin is the atlas's clearest case for why hosting compute is not the same as producing AI. It sits directly on top of AWS, inside Azure, and within reach of a dense corridor that places three providers and thirteen cloud regions within 1,000 km [14, 15, 16]. On the infrastructure side of the scorecard, Dublin matches or exceeds the top-100 reference on proximity, provider diversity, and redundancy. It ranks 16th out of the top 1,000 cities on the bundle index overall.

Yet the city appears in the delivered overlay with only 179 recent AI works and a single top institutional anchor: Dublin City University. It is not a Gi* hot spot. Where Singapore converts infrastructure density into research output at every level, Dublin's cloud footprint outpaces its scholarly footprint. The explanation is structural: Ireland built its tech economy as an English-speaking EU gateway for multinational operations, attracting hyperscaler data centers and corporate headquarters rather than scaling a large indigenous research base. IDA Ireland markets exactly this positioning [17]. CeADAR, Ireland's national Centre for Applied AI [18], Trinity College Dublin [19], and UCD [20] anchor a real but more modestly scaled AI research ecosystem. The scorecard captures that asymmetry clearly: the red bar on urban scale and the moderate score on institutional depth sit alongside near-perfect infrastructure metrics.

Ho Chi Minh City

Distance: 1,096.8 km, Providers within 1,000 km: 0, Cloud regions within 1,500 km: 3, OpenAlex AI works: 373, Bundle score: 61.7/100

Ho Chi Minh City is the atlas's clearest case of a city that beats the simple distance story. It sits 1,096.8 km from the nearest major cloud region (AWS in Singapore), has no providers or cloud regions inside the 1,000 km threshold, and scores a bundle of 61.7, far below Singapore or Dublin. Yet the delivered overlay records 373 recent AI works and two top institutional anchors, above what a pure proximity ranking would imply. It is not a Gi* hot spot, but it is an overperformer relative to raw distance.

Vietnam's national AI strategy for 2021–2030 sets an explicit goal of making the country an innovation and AI development centre in ASEAN and the world [21]. VinAI, one of the country's flagship AI firms, places nearly 200 researchers and engineers across Hanoi and Ho Chi Minh City, reporting 88 top-tier publications — including papers at CVPR, NeurIPS, ICML, and ICLR — in its first three years [22, 23]. VNU-HCM frames itself as a high-quality research and technology hub with AI among its priority areas [24].

The regional map explains why that institutional build-out does not imply isolation. Vung Tau's cable landing stations sit roughly 125 km from the city [25], and VNNIC documents both VNIX exchange points and root-server infrastructure in Ho Chi Minh City itself [26]. The city is farther from hyperscaler regions in kilometres than it is from the wider Southeast Asian network corridor that connects Vietnam into Singapore and the broader region.

Lagos

Distance: 3,842.6 km, Providers within 1,000 km: 0, Cloud regions within 1,500 km: 0, OpenAlex AI works: 0, Bundle score: 27.8/100

Lagos is the atlas's stacked-disadvantage case. It is one of the largest cities in the full 8,000-city frame, but its nearest cloud region in the current package is 3,842.6 km away. It has zero providers and zero cloud regions within 1,000 km, zero observed AI works in the delivered overlay, and no top-institution anchors. Because Lagos is absent from the overlay entirely, it does not receive a Gi* classification at all. The scorecard reads as a large market sitting outside the active compute corridor, and urban scale is the only component where Lagos resembles the top bundle cities.

The mechanism is not just the absence of connectivity. Hyperscaler regions serving Africa remain concentrated in South Africa, leaving West Africa without a local AWS, Azure, or Google Cloud region [14, 15, 27]. Nigeria simultaneously faces a direct infrastructure constraint for compute: the World Bank describes the country's unreliable and often inaccessible power supply as a threat to economic growth [28]. These help explain why local cloud depth has lagged.

Yet Lagos is not disconnected from the internet economy. Submarine Networks records multiple cable landings in Nigeria, including Google's Equiano landing at OADC Lagos [29], and IXPN reports major gains from local peering [30]. Nigeria also remains one of Africa's densest startup markets: Partech's 2025 Africa Tech report records the continent's second-highest equity deal count [31]. The point is not that Lagos lacks demand. It is that bandwidth, startup activity, and urban scale have not yet been matched by local hyperscaler compute.

Conclusion

Singapore, Dublin, Ho Chi Minh City, and Lagos each test the atlas against the complexity of a real urban system. From the details and nuances of each city, we can see that distance alone is insufficient in explaining the story behind accessibility to AI. Proximity to compute matters, but other factors such as provider diversity, regional redundancy, institutional anchors, and urban scale all play an important role as well.We have seen that cities that score well across the full bundle tend to show stronger AI activity, but this is not guaranteed. Cities where only one or two components are present, such as infrastructure without institutions, or demand without compute, tend to show the gaps the atlas is built to make visible.

The strongest contribution of this project is not that it proves a universal causal effect of cloud-region placement. Rather, it unveils the hidden infrastructure layer of AI opportunity, making it visible, measurable, and explainable on in a geospatial context.